TWO TRIANGULATIONS THAT SHARE ONLY THEIR HULL

VAMSHI JANDHYALA

Abstract. Any two triangulations of a finite planar point set share the edges of its convex hull. We show that for every $n \geq 9$ there are n points in general position, with a pentagonal hull, that have two triangulations sharing nothing else. At least five edges must be shared whenever $n \geq 5$, hence $f(n) = 5$ for every $n \geq 9$, where $f(n)$ is the least number of shared edges over all sets of n points. The construction starts from nine explicit points and inserts points one at a time, carrying along a configuration that makes the next insertion possible. Started from a convex polygon, the same construction settles every hull size: n points with h hull vertices admit two triangulations sharing only the hull exactly when $n = h$, or $h \geq 6$, or $h = 5$ and $n \geq 9$. The impossible cases with a pentagonal hull and six to eight points rest on an exhaustive search over order types, which also gives the minimum $4, 5, 6, 6, 6, 5, 5$ for $n = 4, \ldots, 10$.

1. Introduction

How few edges can two triangulations of the same planar point set have in common? The hull edges are forced, but it is not apparent whether anything else must be shared once points lie inside the hull.

Let P be a set of n points in the plane, no three on a line. A *triangulation* of P is a maximal set of pairwise non-crossing segments with endpoints in P; it divides the convex hull of P into triangles, its *faces*. No edge can cross an edge of the convex hull, so every triangulation contains every hull edge, and two triangulations of P always share at least as many edges as the hull has vertices.

Five points in convex position give a first answer. Number them 1 to 5 around the pentagon. The triangulation with diagonals 13 and 14 and the one with diagonals 24 and 25 have no diagonal in common, so they share only the five hull edges. Now put a sixth point inside the pentagon and try to keep this property. A search over all small point sets (Section 6) shows that it cannot be done with six, seven or eight points. With nine it can (Figure 1).

The question was raised on MathOverflow in 2023 [5]. Write

$$f(n) = \min_{|P|=n} \min_{A,B} |A \cap B|,$$

where P ranges over sets of n points in general position and A, B over triangulations of P, each regarded as a set of edges. Contributors to the thread observed that a triangular or quadrilateral hull with an interior point forces at least six shared edges, so that $f(n) \geq 5$ for $n \geq 5$ and five can only occur with a pentagonal hull; this also follows from the forced-edge argument of [4]. Random searches found pairs sharing six edges for $8 \leq n \leq 11$, but none sharing five.

Theorem 1. *For every $n \geq 9$ there is a set of n points in general position with a pentagonal hull and two triangulations whose only common edges are the five hull edges. Hence $f(n) = 5$ for all $n \geq 9$.*

The author posted Theorem 1, with the nine-point example and the insertion argument, as an answer to that question on 30 September 2026 [6]. Theorem 2 below is new here.

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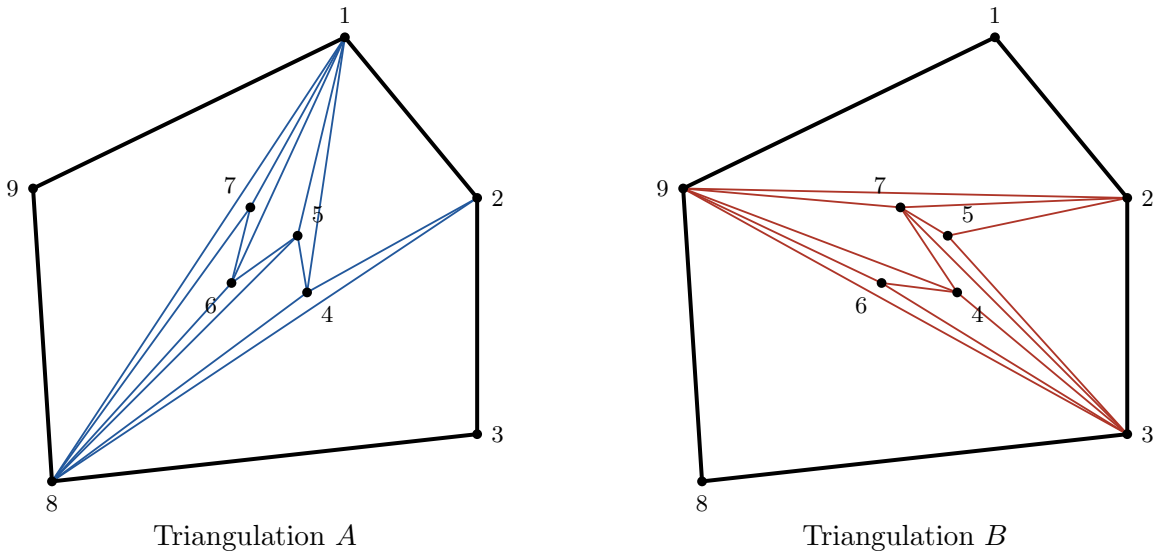


Figure 1. Two triangulations of nine points whose only common edges are the five hull edges (black).

The question is related to geometric thickness. A graph is *doubly linear* if it can be drawn with straight edges on some point set so that its edges split into two non-crossing classes; Hutchinson, Shermer, and Vince [4] showed that such a graph on $n \geq 4$ vertices has at most $6n - 18$ edges. For a point set with a triangular hull, each triangulation has $3n-6$ edges, so two triangulations sharing s edges have a union of $6n - 12 - s$ edges: the shared ones would otherwise be counted twice. The union splits into two non-crossing classes, so the bound says $s \geq 6$ there, and equality would require a doubly linear graph with $6n - 18$ edges. Durocher, Gethner, and Mondal [2] constructed doubly linear graphs with $6n - 19$ edges for every $n \geq 9$, and completing their two layers to triangulations shows $f(n) \leq 7$, as Alex Ravsky observed in the thread [5]. Theorem 1 does not bear on whether $6n - 18$ edges are possible. That question needs a triangular hull, and with a pentagonal hull the union of the two triangulations has only $2(3n - 8) - 5 = 6n - 21$ edges.

The question also makes sense for each hull size separately. García, Hurtado, Korman, Matos, Saumell, Silveira, Tejel, and Tóth [3] studied *biplane* graphs on a fixed point set, those whose edges split into two non-crossing classes. A maximal biplane graph on P is the union of two triangulations [3, Lemma 1], so if the hull of P has h vertices it has at most $2(3n - 3 - h) - h = 6n - 3h - 6$ edges, with equality exactly when the two triangulations share only the hull. For the nine points of Figure 1, with $h = 5$, each triangulation has $3 \cdot 9 - 3 - 5 = 19$ edges and their union has $2 \cdot 19 - 5 = 33 = 6 \cdot 9 - 3 \cdot 5 - 6$. The second theorem says when that happens.

Theorem 2. *Let $3 \leq h \leq n$. There is a set of n points in general position with h hull vertices and two triangulations whose only common edges are the h hull edges if and only if $n = h$, or $h \geq 6$, or $h = 5$ and $n \geq 9$.*

The nonexistence assertion for a pentagonal hull with six to eight points is computer-assisted (Section 6); all other cases have geometric proofs. A triangular or quadrilateral hull with a point inside always forces extra shared edges. For every $h \geq 6$ and $n \geq h$, some point set attains hull-only sharing. The pentagon of the opening example is the only hull size with a gap.

Section 2 proves the lower bound. Section 3 gives the nine-point example, and Sections 4 and 5 show how to add points to it. Section 6 explains why the answer changes at nine, and Section 7 treats the other hull sizes and proves Theorem 2.

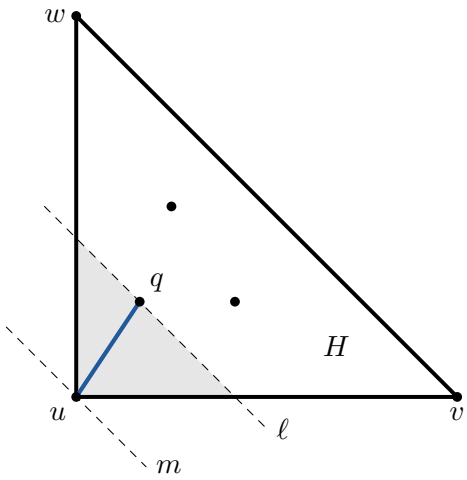


Figure 2. Lemma 3 for a triangular hull. The line m through u is parallel to vw; q is the point nearest to it, and ℓ is the parallel through q. The shaded strip between m and ℓ contains no point of the set, so no segment crosses uq (thick).

2. The Lower Bound

The lower bound comes from finding edges that neither triangulation can avoid. A segment between two points of P is *unavoidable* if every triangulation of P contains it. Hull edges are unavoidable, and so is any segment that no other segment crosses. For an example, take the triangle $u = (0,0)$, $v = (12,0)$, $w = (0,12)$ with the three interior points $(2,3)$, $(5,3)$ and $(3,6)$ (Figure 2). Slide a line parallel to vw in from u. The first point it meets is $q = (2,3)$, and the strip it has swept contains no other point, so no segment between two points of the set can cross uq. The lemma and corollary below are a version of the forced edges of Hutchinson, Shermer, and Vince [4, Lemma 5 and the proof of Theorem 6]; we include the short proofs to keep the paper self-contained.

Lemma 3. *Let u be a hull vertex of P and m a line through u that has all other points of P strictly on one side. If q is a point of $P \setminus \{u\}$ nearest to m, then uq is unavoidable.*

Proof. Let ℓ be the line through q parallel to m, and H the closed half-plane bounded by ℓ that does not contain u. All points of P other than u lie in H. Every point of the open segment uq lies strictly between m and ℓ, hence outside H. A segment crossing uq does so at such a point, so it is not contained in H; but a segment with both endpoints in H lies in H, and a segment with an endpoint at u does not cross uq. So no segment crosses uq. □

Corollary 4. *If P has an interior point and its hull has at most four vertices, two triangulations of P share at least six edges. Hence if $|P| \geq 5$, two triangulations of P share at least five edges, and if they share exactly five then the hull of P is a pentagon.*

Proof. Suppose first that P has an interior point and its hull has at most four vertices. If the hull is a triangle uvw, apply Lemma 3 at u with m parallel to vw; interior points are closer to m than v and w are, so the resulting segment is interior. Doing the same at v and at w gives three interior unavoidable segments, which are distinct because their hull endpoints differ. With the hull edges that makes six. If the hull is a quadrilateral $uvwx$, one of the triangles uvx and wvx contains an interior point; if it is uvx, apply the lemma at u with m parallel to vx, and otherwise at w. This gives an interior unavoidable segment ending at u or w. The diagonal uw gives, in the same way, one ending at v or x. With the four hull edges that makes six.

For the second statement, a hull with at least five vertices gives five shared edges on its own. A hull with at most four vertices and $|P| \geq 5$ leaves an interior point, and then six edges are shared. □

3. Nine Points

Number the points as follows.

i	1	2	3	4	5	6	7	8	9
x_i	33	47	47	29	28	21	23	2	0
y_i	47	30	5	20	26	21	29	0	31

Write ij for the segment from point i to point j. The hull is the pentagon $1, 2, 3, 8, 9$, with edges $12, 23, 38, 89, 91$. Figure 1 shows the triangulations

$$A = \text{hull} \cup \{14, 15, 16, 17, 18, 24, 28, 45, 48, 56, 58, 67, 68, 78\},$$
$$B = \text{hull} \cup \{25, 27, 29, 34, 35, 36, 37, 39, 46, 47, 49, 57, 69, 79\}.$$

The two sets of interior edges are disjoint. One checks from the coordinates that no three points are collinear and that no two edges of A, or of B, cross; the ancillary file `check_paper.py` does this in exact arithmetic. Each of A and B then has 19 pairwise non-crossing edges, and $19 = 3 \cdot 9 - 3 - 5$ is the number of edges of any triangulation of nine points with five on the hull (Euler's formula), so both are triangulations. Hence $f(9) \leq 5$.

4. Adding a Point

We want to preserve the property of the nine-point example as points are inserted. Call two triangulations of the same point set *hull-disjoint* if they share only the hull edges.

To add a point without creating a shared edge, it is enough to place it inside a face of A and a face of B that have no vertex in common, and join it to the vertices of each.

Definition 5. Let A and B be triangulations of P. A face α of A and a face β of B form a *good pair* if their interiors intersect and they have no vertex in common.

For example, in Figure 1 the face 156 of A and the face 479 of B form a good pair; the point $(24, 24)$ lies inside both.

Lemma 6 (Insertion)**.** *Let A and B be hull-disjoint triangulations of P with a good pair (α, β), and let $p \in \text{int}(\alpha) \cap \text{int}(\beta)$ lie on no line through two points of P. Let A' be A with p joined to the three vertices of α, and B' be B with p joined to the three vertices of β. Then A' and B' are hull-disjoint triangulations of $P \cup \{p\}$.*

Proof. As Figure 3 illustrates, joining a point inside a face to its three vertices cuts that face into three triangles and leaves everything else unchanged, so A' and B' are triangulations; the hull is the same as before. The old edges of A' and B' are shared only on the hull. The new edges of A' join p to the vertices of α, and those of B' join p to the vertices of β; as α and β have no common vertex, no new edge is shared. A suitable p exists because $\text{int}(\alpha) \cap \text{int}(\beta)$ is a nonempty open set and only finitely many lines are excluded. □

5. The Good Pair Returns

Lemma 6 handles one insertion; to repeat it, a new good pair is needed after each step. Look again at Figure 1. The side 56 of the face 156 of A passes through the faces 357, 347, 479 and 469 of B, in this order from 5 to 6. The face 479 shares no vertex with 156, so it forms a good pair with 156 and can receive the next point. The face 347 contains neither 5 nor 6, and it will still lie along the side 56 after that point is inserted, ready to form the next good pair. Carrying both faces along one side is the whole idea.

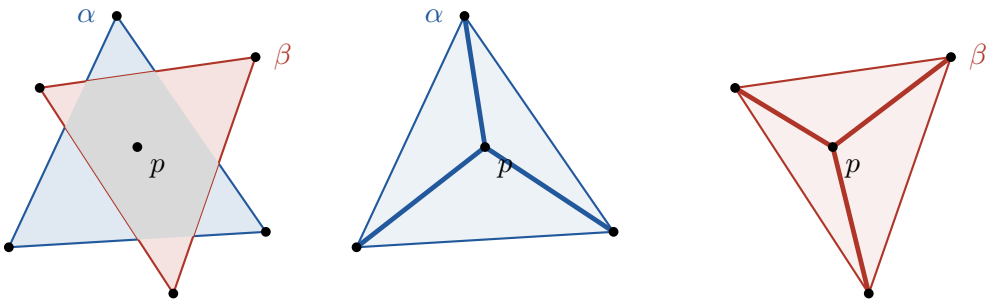


Figure 3. Lemma 6. The point p lies in the overlap of α and β (shaded). The three thick blue edges are added to A and the three thick red edges to B.

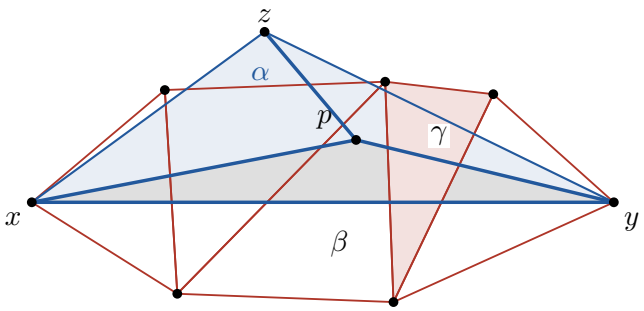


Figure 4. Lemma 8. The side xy of $\alpha = xyz$ passes through the faces β and γ of B (red; faces of B away from xy are omitted, and β is drawn before it is subdivided). After p is inserted in $\alpha \cap \beta$, the face pxy of A' (grey), the face γ (shaded) and a piece of β form the next chain.

Definition 7. Let A and B be triangulations of P. A face $\alpha = xyz$ of A and two different faces β and γ of B form a *chain* $(\alpha; \beta, \gamma)$ along the side xy if the open segment xy meets the interiors of both β and γ, β has no vertex in common with α, and γ has neither x nor y as a vertex.

So $(156; 479, 347)$ is a chain along 56.

A chain contains a good pair. Near a point of the open side xy inside β, the face α contains a half-disc bounded by xy, so the interiors of α and β meet, and they have no common vertex. So Lemma 6 applies to (α, β), and the chain survives the insertion.

Lemma 8 (Regeneration)**.** *Let A and B be hull-disjoint triangulations of P with a chain $(\alpha; \beta, \gamma)$ along the side xy of α, and let A', B' be obtained from the good pair (α, β) as in Lemma 6. Then some face β' of B' inside β meets the open segment xy, and $(pxy; \gamma, \beta')$ is a chain of A' and B' along xy.*

Proof. The face pxy of A' has side xy. The face γ is still a face of B', since only β was subdivided, and it still meets the open segment xy. Inserting p cuts β into three faces along the segments from p to the vertices of β. Each of these segments meets the line xy at most once, because p is on no line through two points of P. The open segment xy meets $\operatorname{int}(\beta)$ in an open interval, so some point of that interval avoids the three segments and lies inside one of the three faces; call it β'. It remains to check the vertices. The vertices of γ are points of P other than x and y, and $p \notin P$, so γ has no vertex in common with pxy. The vertices of β' are p and two vertices of β, which are neither x nor y. Finally $\beta' \subseteq \beta$ is different from γ. □

In the example, inserting $p = (24, 24)$ in $156 \cap 479$ gives the chain $(56p; 347, \beta')$, where β' is the piece $47p$ of 479. In particular $(56p, 347)$ is the next good pair.

Proof of Theorem 1. The triangulations of Section 3 are hull-disjoint and have the chain (156; 479, 347). By Lemmas 6 and 8, points can be added one at a time, keeping the pentagonal hull, hull-disjoint triangulations and a chain. This gives $f(n) \leq 5$ for every $n \geq 9$, and Corollary 4 gives $f(n) \geq 5$. □

At each step the point p can be chosen rational, so for each n the construction yields integer coordinates after clearing denominators.

6. Why Nine

Aichholzer, Aurenhammer, and Krasser [1] listed all order types of up to ten points in general position. Two point sets have the same order type if some bijection between them preserves the orientation of every triple or reverses all of them, so a set and its mirror image count once; they then have the same crossing pairs of segments and hence the same triangulations. There are 3,315 order types of eight points, 158,817 of nine and 14,309,547 of ten.

For a single point set the minimum is a finite computation. Two triangulations of P have $2(3n-3-h)$ edges counted with multiplicity, where h is the number of hull vertices, so they share $2(3n-3-h) - |A \cup B|$ edges; and $A \cup B$ is a set of segments that splits into two non-crossing classes, namely A and $B \setminus A$. So $|A \cup B|$ is at most the size of a largest such set. Conversely, let U be a largest set of segments that splits into two non-crossing classes U_1 and U_2. A non-crossing set of segments extends to a triangulation, so we may choose triangulations $A \supseteq U_1$ and $B \supseteq U_2$. This extension cannot enlarge the union: $A \cup B$ contains U, and it again splits into the two non-crossing classes A and $B \setminus A$, so by the maximality of U it has no more segments than U. Hence $A \cup B = U$, and the largest union of two triangulations of P is exactly a largest set of segments that splits into two non-crossing classes. So the minimum number of shared edges is

$$6n - 6 - 2h - \binom{n}{2} + \operatorname{oct}(P),$$

where $\operatorname{oct}(P)$ is the least number of segments that must be removed from the crossing graph of all $\binom{n}{2}$ segments to make it bipartite. In words, $\binom{n}{2} - \operatorname{oct}(P)$ is the size of a largest set of segments that splits into two non-crossing classes, which we have just seen is the largest possible union $A \cup B$; subtracting it from the $2(3n-3-h)$ edges of A and B leaves the overlap. For a convex pentagon the five hull edges cross nothing and the five diagonals cross in a single 5-cycle, so oct = 1 and the formula gives $30 - 6 - 10 - 10 + 1 = 5$: two triangulations of a pentagon can share only the hull.

The crossing graph has one vertex for each segment and an edge for each crossing pair. The search tests whether at most k of its vertices can be deleted to leave a bipartite graph. If an odd cycle remains, every successful deletion set must contain a vertex of that cycle; branching over all its vertices is therefore exhaustive. A branch stops when the remaining graph is bipartite or its deletion budget is exhausted. Vertex-disjoint odd cycles give a lower bound on the deletions needed and allow branches to be pruned. All orientation tests and graph operations use exact integer arithmetic.

For the computer-dependent part of Theorem 2, the database has 3, 22 and 570 pentagonal-hull order types on six, seven and eight points respectively. None permits only five shared edges. The ancillary files give commands, expected counts and hashes of the input files, so this small negative result can be reproduced separately from the larger searches.

Evaluating the formula for every order type gives, for point sets in general position,

n	4	5	6	7	8	9	10
$\min \lvert A \cap B \rvert$	4	5	6	6	6	5	5

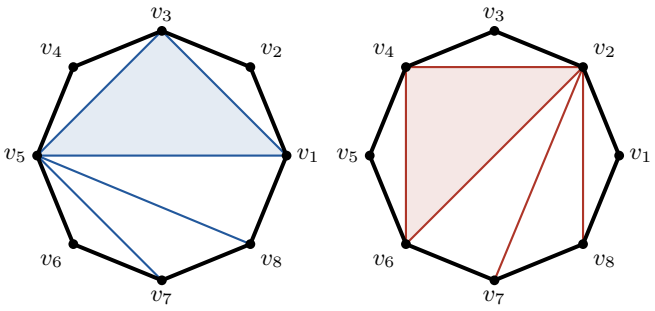


Figure 5. Lemma 9 for $h = 8$. The side v_1v_3 of the shaded face $v_1v_3v_5$ of A passes through the faces $v_2v_4v_6$ (shaded) and $v_2v_6v_7$ of B. For $h = 6$ the fans are empty.

So the minimum rises to six and returns to five at $n = 9$. Exactly seven order types of nine points and 734 of ten attain five, all with pentagonal hulls; the set of Section 3 is one of the seven. For a triangular hull the minimum is seven when $n = 9$ and when $n = 10$, which means that the largest doubly linear graphs on nine or ten points in general position have $6n - 19$ edges. The ancillary files contain the search program, its output, and an independent check of the seven nine-point sets.

7. Other Hulls

To start the construction of Section 5 with another hull size, we need hull-disjoint triangulations with a chain. Convex polygons provide them.

Lemma 9 (Convex seed)**.** *Let $v_1, \dots, v_h$ be the vertices of a convex polygon in order around it, with $h \geq 6$, and let*

$$A = \mathit{hull} \cup \{v_1v_3,\ v_3v_5,\ v_1v_5\} \cup \{v_5v_k : 7 \leq k \leq h\},$$
$$B = \mathit{hull} \cup \{v_2v_4,\ v_4v_6,\ v_2v_6\} \cup \{v_2v_k : 7 \leq k \leq h\}.$$

Then A and B are hull-disjoint triangulations of $\{v_1, \dots, v_h\}$. If $h \geq 7$, the face $v_1v_3v_5$ of A and the faces $v_2v_4v_6$ and $v_2v_6v_7$ of B form a chain along v_1v_3.

Proof. The triangle $v_1v_3v_5$ cuts the polygon into itself, the ears $v_1v_2v_3$ and $v_3v_4v_5$, and the convex polygon $v_5v_6 \cdots v_hv_1$, which the segments v_5v_k triangulate as a fan from v_5. So A is a triangulation. The same argument, with every index increased by one, shows that B is one: its pieces are $v_2v_4v_6$, the ears $v_2v_3v_4$ and $v_4v_5v_6$, and the polygon $v_6 \cdots v_hv_1v_2$ fanned from v_2. Every interior edge of B has an endpoint among v_2, v_4, v_6, and no interior edge of A does, so A and B share only the hull. Now let $h \geq 7$. The segment v_1v_3 separates v_2 from the other vertices, so it crosses every segment from v_2 to another vertex except the two hull edges at v_2. In particular it crosses the sides v_2v_4 and v_2v_6 of $v_2v_4v_6$, and the sides v_2v_6 and v_2v_7 of $v_2v_6v_7$, and so meets the interiors of both faces. The face $v_2v_4v_6$ has no vertex in common with $v_1v_3v_5$, and $v_2v_6v_7$ contains neither v_1 nor v_3. □

For $h = 6$ these triangulations have no chain. Every side of a face of A that is not a hull edge is a side of $v_1v_3v_5$. The side v_1v_3 passes through only three faces of B, namely $v_6v_1v_2$, $v_2v_4v_6$ and $v_2v_3v_4$, and the outer two contain v_1 or v_3; by symmetry the same holds for v_3v_5 and v_5v_1. So the hexagonal case needs a different start, and seven points suffice.

Number the points as follows.

i	1	2	3	4	5	6	7
x_i	1	0	1	4	5	6	2
y_i	2	0	0	1	2	6	1

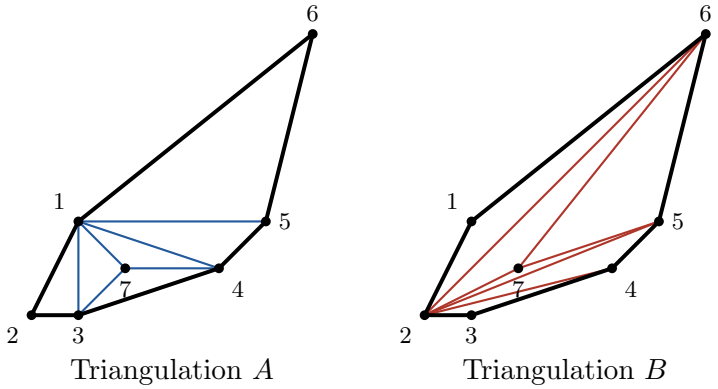


Figure 6. Two triangulations of seven points with a hexagonal hull, sharing only the six hull edges (black).

The hull is the hexagon $1, 2, 3, 4, 5, 6$, and point 7 lies inside it. Figure 6 shows the triangulations

$$A = \text{hull} \cup \{13, 14, 15, 17, 37, 47\},$$
$$B = \text{hull} \cup \{24, 25, 26, 27, 57, 67\}.$$

As in Section 3, one checks from the coordinates that no three points are collinear and that no two edges of A, or of B, cross. Each has $12 = 3 \cdot 7 - 3 - 6$ edges, so both are triangulations, and their interior edges are disjoint. The side 13 of the face 137 of A passes through the faces 245 and 267 of B (the point $(6/5, 2/5)$ lies inside 137 and 245). The face 245 shares no vertex with 137, and 267 contains neither 1 nor 3, so $(137; 245, 267)$ is a chain along 13.

Proof of Theorem 2. Suppose first that $n = h$, so the points are in convex position. For $h = 3$ the only triangulation has no interior edge. For $h = 4$ the two diagonals give two triangulations sharing only the hull, and for $h = 5$ so do the fans $\{v_1v_3, v_1v_4\}$ and $\{v_2v_4, v_2v_5\}$. For $h \geq 6$, Lemma 9 applies.

Now let $n > h$, so that P has an interior point. If $h \leq 4$, Corollary 4 gives at least six shared edges, which is more than h. If $h = 5$, the table in Section 6 shows that two triangulations of six, seven or eight points share at least six edges, and Theorem 1 covers every $n \geq 9$. If $h \geq 7$, start from the chain of Lemma 9. Lemmas 6 and 8 add interior points one at a time, keeping the hull, hull-disjoint triangulations and a chain, and every $n > h$ is reached. If $h = 6$, start in the same way from the seven points above. □

A longer argument, included in the Lean formalisation described in the acknowledgements, shows that once $|P| \geq 7$ every good pair of hull-disjoint triangulations is followed by another after an insertion, with no chain needed; the hexagon of the convex seed is where this fails. The part of the proof for a pentagonal hull with six to eight points is the only one that rests on the computer search; a proof by hand would be welcome. The enumeration agrees with Theorem 2 throughout its range: for every $n \leq 9$ and every h, some order type with n points and h hull vertices has two triangulations sharing only the hull exactly when the theorem says so.

We end with two further questions. Is the minimum for a triangular hull equal to seven for every $n \geq 9$? And can a set of six, seven, or eight points with collinear triples, which the enumeration does not include, have two triangulations sharing only five edges?

Acknowledgements

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Claude Fable 5.1 (Anthropic) were used to search for examples, to write and run the verification code, to develop and check the proofs, and to draft the manuscript. The author is responsible for the content. The ancillary files of this article contain an exact checker for the examples and small claims, including those of Section 7, the enumeration code, and coordinates and both triangulations for $9 \leq n \leq 17$. Theorems 1 and 2 and the lemmas they use have also been formalised in the Lean 4 proof assistant, with Lemma 9 checked for points on a parabola, which is the case the proof of Theorem 2 uses. The formalisation takes the computer search of Section 6 as a stated hypothesis for a pentagonal hull with six to eight points, and the supplementary archive includes the Lean sources, pinned dependency versions, build instructions and a driver that prints the theorem statements and their axiom dependencies. OpenAI ChatGPT assisted with the subsequent exposition and reproducibility revision. The certificates are also available at [7].

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[7] V. Jandhyala, *Two triangulations that share only their hull*: certificates, enumeration code and Lean sources, `https://github.com/jvvk/mathematics/tree/main/two-triangulations-hull`.

London, United Kingdom

Email address: `vamshi.jandhyala@gmail.com`